Oxford College of Emory University

Chapter 13: Fiscal Policy

Government Purchases, Taxes, and Transfers

ECON_OX 112 · Principles of Macroeconomics

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The Federal Budget: FY2025

Federal government only · October 2024–September 2025 · Billions of dollars

Revenues	$ billions
Individual income taxes	2,656
Payroll taxes	1,748
Corporate income taxes	452
Customs duties (tariffs)	195
Excise taxes	106
Estate and gift taxes	29
Federal Reserve remittances	5
Miscellaneous fees and fines	42
Total revenues	**5,235**

Outlays	$ billions
Social Security	1,575
Medicare (net)	988
Medicaid	668
Income security programs	397
Other mandatory (net)	539
Defense discretionary	893
Nondefense discretionary	980
Net interest	970
Total outlays	**7,010**

Budget balance: $5,235 − $7,010 = −$1,775 billion (deficit)

A financing shortfall—not a measure of the effect on GDP.

Source: CBO, FY2025 Federal Budget; Revenues; 2026 Outlook. Components may not sum because of rounding. Medicare and other mandatory outlays are net of offsets.

What Does Federal Spending Pay For?

> **Public services**
>
> National defense, federal courts, and law enforcement.

Source: CBO, Federal Budget and spending breakdowns, FY2025

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Retirement and disability benefits, food assistance, and refundable tax credits.

Interest on federal debt

Payments to lenders; these do not buy current goods or services.

Source: CBO, Federal Budget and spending breakdowns, FY2025

Roadmap

Policy goals and instruments

Purchases, household taxes, and transfers

Demand and policy limits

Expansion, contraction, criticisms, and delays

The multiplier

Initial spending and successive consumption rounds

Automatic stabilization

Existing rules versus new policy decisions

Fiscal Policy and Stabilization

Definition

Government decisions about purchases, taxes, and transfers.

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Stabilization goal

Support output and employment in a recession; restrain excess demand in a boom.

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Policy direction

Expansionary policy raises AD; contractionary policy reduces AD.

Public services and redistribution are also fiscal goals.

Fiscal Instruments

Instrument	Example	Immediate channel
Purchases (G)	Road maintenance	A purchase of current output
Transfers (TR)	Unemployment benefits	Household disposable income
Household taxes (T)	Personal income taxes	Household disposable income

Transfers are not G. Public infrastructure investment is G; I is private investment.

GDP accounting: BEA

Fiscal Policy and Aggregate Expenditure

$$Y = \mathbf{C} + \mathbf{I} + \mathbf{G} + \mathbf{NX}$$

C: Consumption

Household taxes and transfers change disposable income.

Tax cuts or higher transfers can raise C.

I: Private investment

G: Government purchases

NX: Net exports

Fiscal Policy and Aggregate Expenditure

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Fiscal Policy and Aggregate Expenditure

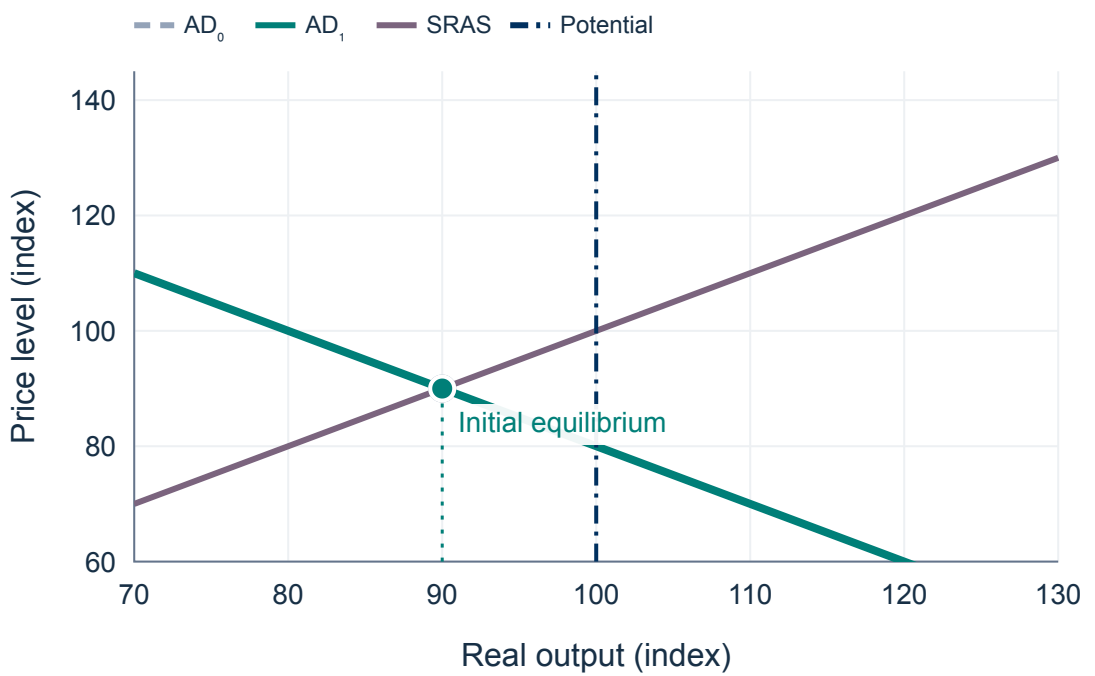
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NX: Net exports	Higher domestic spending can increase imports. Imports reduce NX, other things equal.

Expansionary Fiscal Policy

Starting output = 90 · Potential output = 100

Expansionary · AD increases



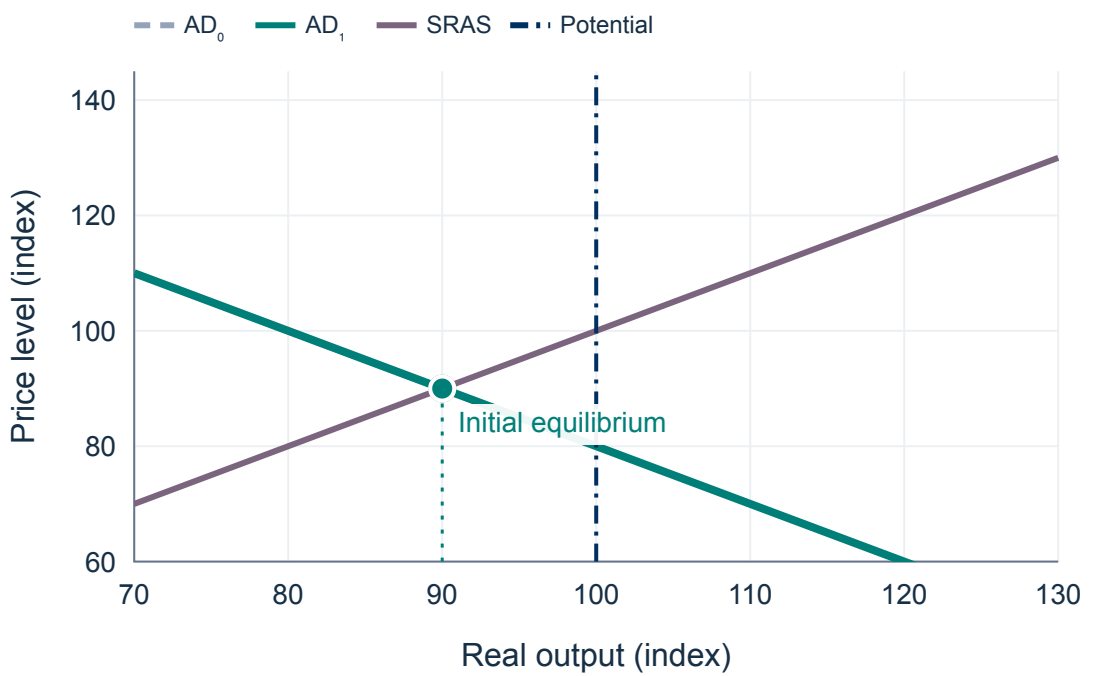
Recessionary gap

Output is below potential.

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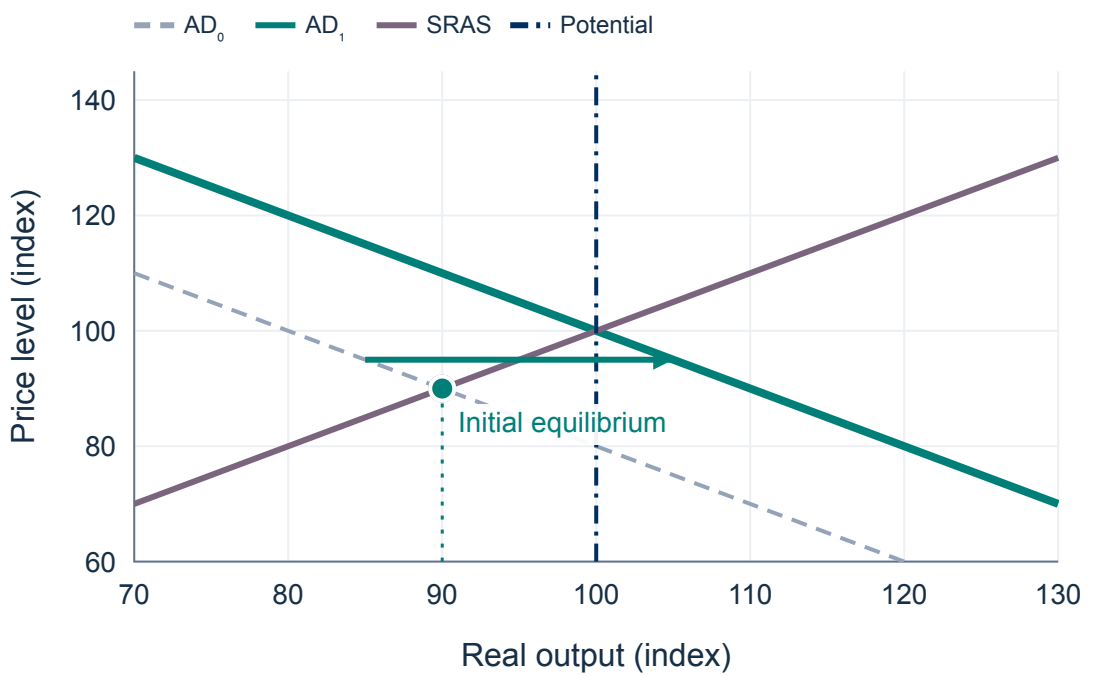
Policy change

Government purchases increase.

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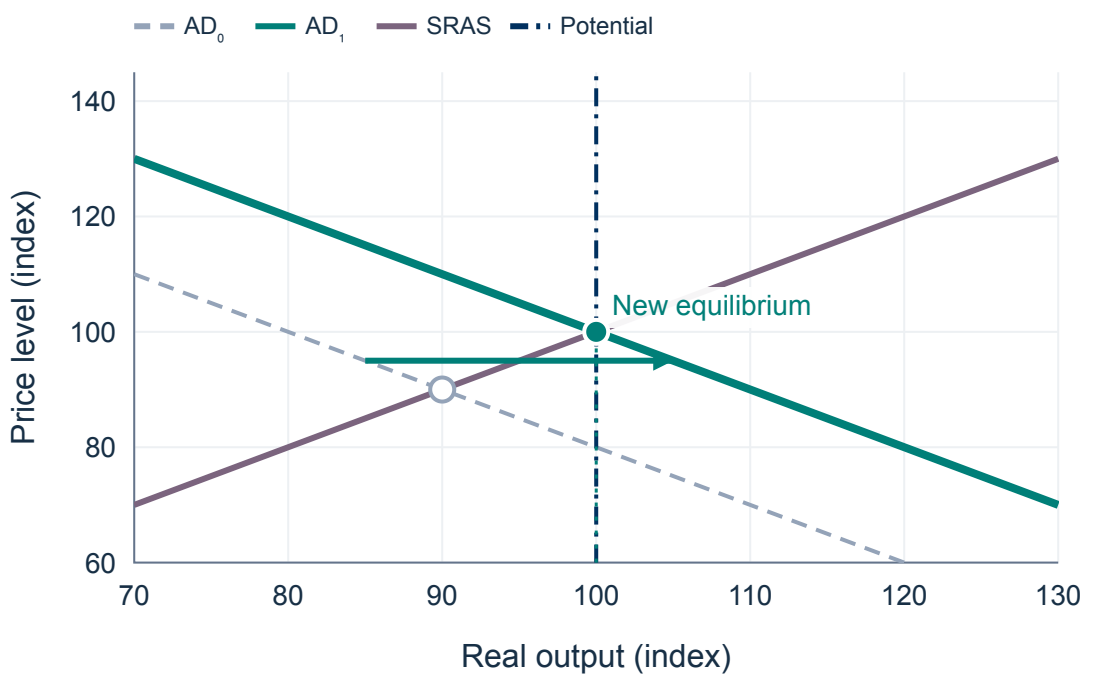
Demand response

AD shifts right; production rises.

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New equilibrium

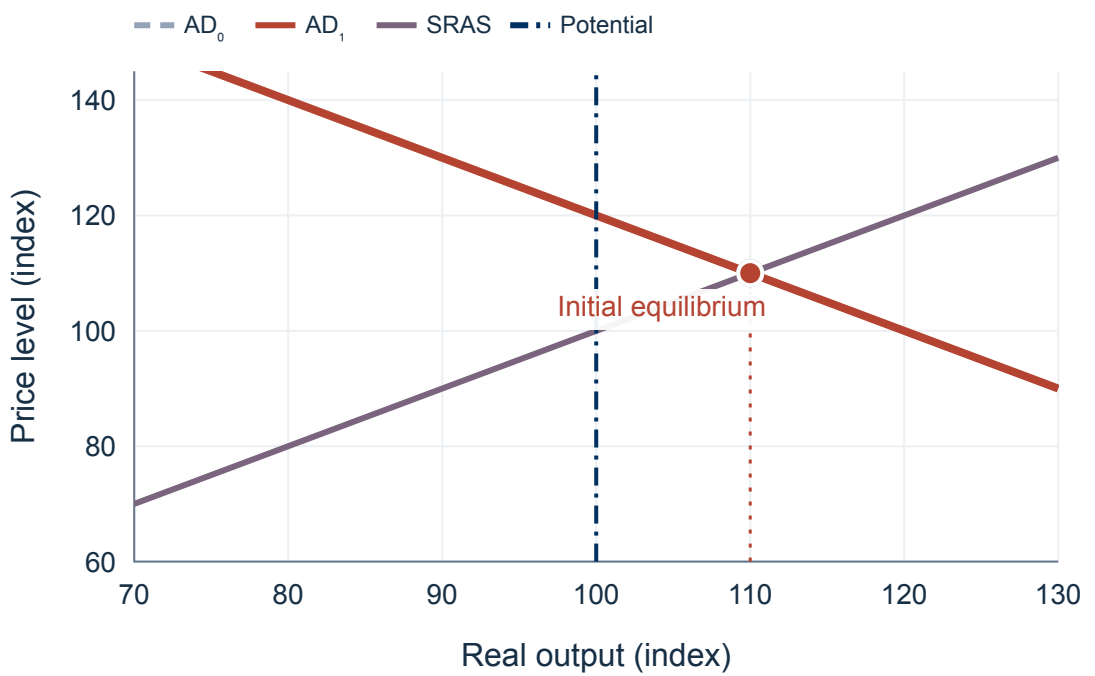
Output reaches 100; the price level rises.

Expansionary alternatives: G ↑, household T ↓, or TR ↑.

Contractionary Fiscal Policy

Starting output = 110 · Potential output = 100

Contractionary · AD decreases



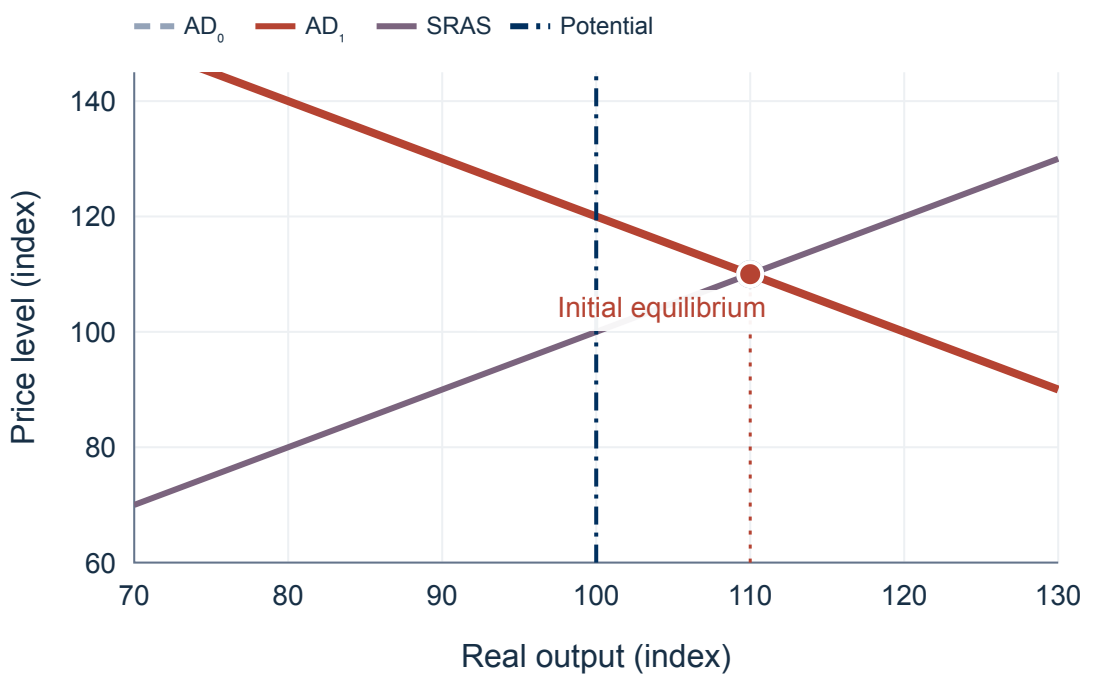
Inflationary gap

Output is above potential.

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Contractionary · AD decreases



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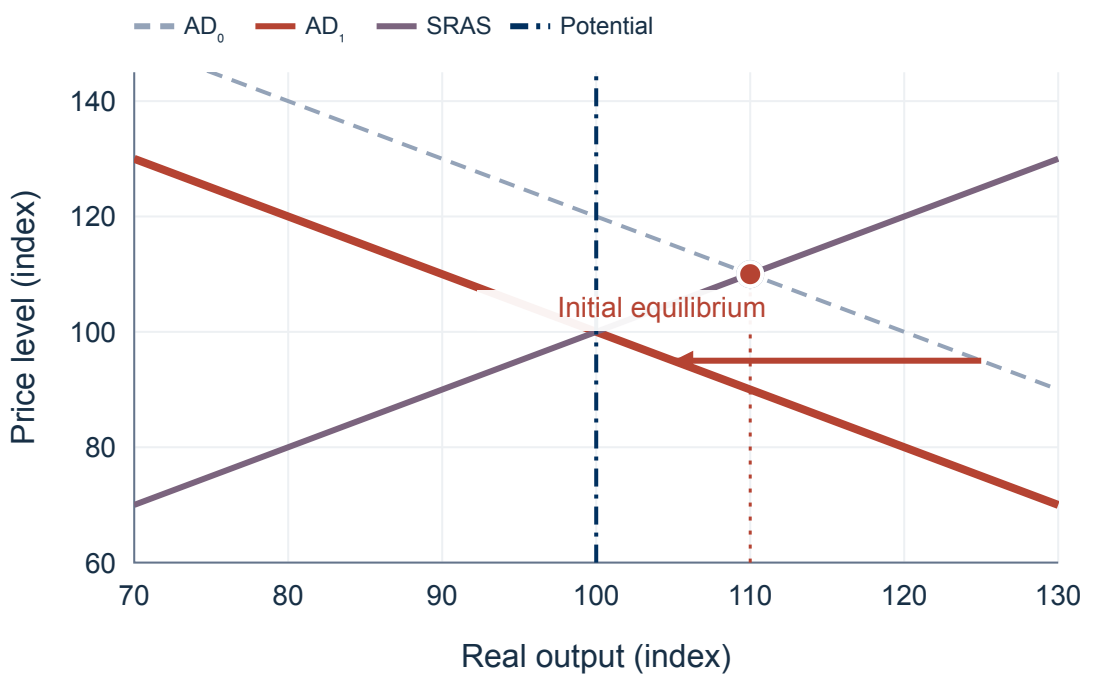
Policy change

Government purchases decrease.

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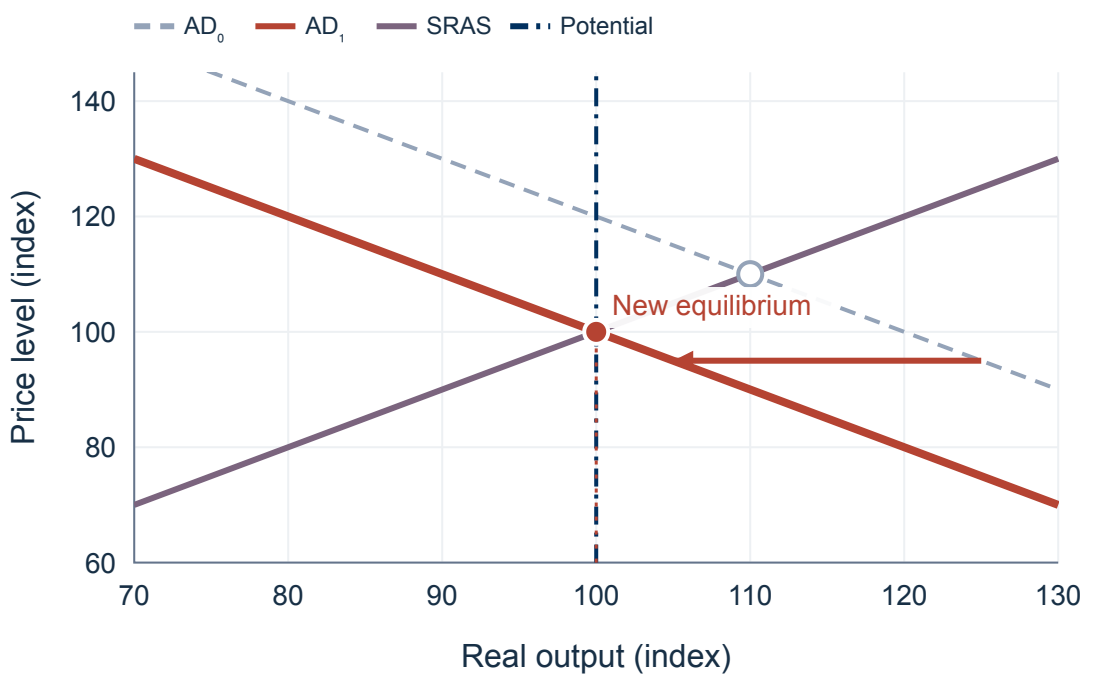
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AD shifts left; production falls.

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Policy change

Government purchases decrease.

Demand response

AD shifts left; production falls.

New equilibrium

Output returns to 100; the price level falls.

Contractionary alternatives: G ↓, household T ↑, or TR ↓.

Fiscal Policy Checkpoint

Identify the policy direction and the immediate spending channel.

A. Output is below potential; government repairs more roads.

B. Output is above potential; household income taxes rise.

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Contractionary. Disposable income and consumption fall.

Three Criticisms of Fiscal Expansion

Criticism	Mechanism	Qualification
Investment crowding out	Borrowing can raise interest rates and reduce I.	The size depends on economic and monetary conditions.
Consumption offset		
Ricardian equivalence		

An offset can weaken a policy without eliminating its effect.

References: CBO fiscal-policy framework; CRS fiscal-policy review

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Consumption offset	Current tax financing can reduce disposable income and C.	The reduction need not cancel the extra G.
Ricardian equivalence	A tax cut may be saved for expected future taxes.	A conditional tax-timing argument, holding spending fixed.

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Fiscal Policy Lags: The 2008–2009 Response

Recognition

Diagnose the shock.

15 Sep 2008

Lehman bankruptcy

26 Jan 2009

Recovery bill introduced

17 Feb 2009

ARRA signed

May 2009

$250 benefit payments

Authorization and delivery occur at different times.

Sources: Federal Reserve History; Congress.gov (H.R. 1); Social Security Administration

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The Spending Multiplier

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Recipients consume part and save part of the extra income.

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Further rounds become smaller

New consumption creates more income; saving limits the chain.

The output change combines the first spending change and the later rounds.

Government Purchases: Successive Spending Rounds

ΔG = +$100 billion · MPC = 0.8 · Fixed price level

Government purchases

+$100 billion

The initial purchase is income for workers and suppliers.

First spending

$100 billion

Fixed taxes and transfers; no imports or investment offsets.

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0.8 × $100 = $80 billion

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$64 billion + $51.2 billion + …

Total ΔY = $500 billion $k_G = 1/(1 - 0.8) = 5$

Fixed taxes and transfers; no imports or investment offsets.

A Household Tax Cut

Household taxes fall by $100 billion · MPC = 0.8

> **Disposable income rises**
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> $\Delta T = -\$100$ billion; $\Delta Y_D = +\$100$ billion.

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Initially, consumption rises $80 billion and saving rises $20 billion.

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$\Delta Y = \$80 \text{ billion} \times 5 = +\$400$ billion.

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The tax multiplier has a negative sign

$k_T = -4$, so $\Delta Y = (-4) \times (-\$100 \text{ billion}) = +\$400$ billion.

Purchases, Tax Cuts, and Transfers

MPC = 0.8 · Change one instrument at a time

Expansionary change	First channel	Initial expenditure
G increases by $100 billion	Government purchases	+$100 billion in G
Household T falls by $100 billion		
TR increases by $100 billion		

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TR increases by $100 billion	Disposable income +$100 billion	+$80 billion in C

Transfers affect GDP when recipients spend them; the payment itself is not G.

Fiscal Multipliers

MPC = 0.8 · Fixed price level · One policy change at a time

Expansionary change	First spending	Multiplier	Equilibrium ΔY
ΔG = +$100 billion	+$100 billion	$k_G = 5$	+$500 billion
ΔT = −$100 billion			
ΔTR = +$100 billion			

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$k_G = 1/(1 - MPC)$ $k_T = -MPC/(1 - MPC)$ $k_{TR} = MPC/(1 - MPC)$

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A $500 billion gap: ΔG = $500 billion ÷ 5 = $100 billion.

Recipients and Policy Design

An additional $100 of disposable income

Recipient	MPC	Initial consumption
Household A	0.8	$80
Household B	0.4	

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Consider spending, delivery speed, distribution, and the value of public purchases.

Income Taxes and the Multiplier

MPC = 0.8 · Income-tax rate t = 25% · Fixed price level

Additional output

→ →

$\Delta Y = \$100$

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Additional output

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$\rightarrow$

Taxes and disposable income

$\Delta T = \$25 \quad \Delta Y_D = \$75$

$\rightarrow$

Income Taxes and the Multiplier

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Additional output		Taxes and disposable income		Additional consumption
$\Delta Y = \$100$	$\rightarrow$	$\Delta T = \$25$ $\Delta Y_D = \$75$	$\rightarrow$	$\Delta C = 0.8 \times \$75 = \$60$

Only \$60 reaches the next spending round, compared with \$80 under fixed taxes.

MPC remains 0.8; MPS remains 0.2.

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$k_G = 1 / [1 - MPC(1 - t)]$

$k_G = 1 / [1 - 0.8(0.75)] =$ **2.5**, compared with **5** under fixed taxes.

Automatic Stabilizers

Income falls

Tax payments fall under existing income-tax rules.

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Consumption is cushioned

Disposable income falls by less than it otherwise would.

No new law is needed for these responses.

Fiscal Policy: Final Checkpoint

Explain the mechanism, not just the label.

A. Income falls; the tax schedule stays unchanged.

B. A new law raises weekly unemployment benefits.

C. With MPC = 0.8, compare a $100 purchase and a $100 tax cut.

Fiscal Policy: Final Checkpoint

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Purchases: $\Delta Y = \$500$. Tax cut: $\Delta Y = \$400$; the first round is smaller.